Taewon Park

PhD Candidate · Electrical Engineering
Stanford University
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Research Interests: Integrated photonics, quantum optics, quantum sensing, nonlinear optics

Education

Stanford University, Stanford, CA, USA
PhD in Electrical Engineering 2020.9 – Present
Advisor: Amir Safavi-Naeini (Applied Physics Department)

Stanford University, Stanford, CA, USA
MS in Electrical Engineering 2024.6

Korea Advanced Institute of Science and Technology (KAIST), Daejeon, South Korea
BS in Electrical Engineering 2013.3 – 2020.2
Advanced Major Program · Graduation with Honors · Undergraduate research advisor: Kyoungsik Yu

Journal Publications

† denotes co-first authorship

[1] Devin J Dean†, **Taewon Park**†, Hubert S Stokowski, Luke Qi, Sam Robison, Alexander Y Hwang, Jason F Herrmann, Martin M Fejer, Amir H Safavi-Naeini. "Low-power integrated optical amplification through second-harmonic resonance." *Nature* 649(8099), 1159–1164 (2026).

[2] **Taewon Park**, Hubert S Stokowski, Vahid Ansari, Samuel Gyger, Kevin KS Multani, Oguz Tolga Celik, Alexander Y Hwang, Devin J Dean, Felix Mayor, Timothy P McKenna, Martin M Fejer, Amir H Safavi-Naeini. "Single-mode squeezed-light generation and tomography with an integrated optical parametric oscillator." *Science Advances* 10(11), eadl1814 (2024).

[3] **Taewon Park**, Hubert S Stokowski, Vahid Ansari, Timothy P McKenna, Alexander Y Hwang, Martin M Fejer, Amir H Safavi-Naeini. "High-efficiency second harmonic generation of blue light on thin-film lithium niobate." *Optics Letters* 47(11), 2706–2709 (2022).

[4] **Taewon Park**, Youngjae Jeong, Kyoungsik Yu. "Cascaded optical resonator-based programmable photonic integrated circuits." *Optics Express* 29(3), 4645–4660 (2021).

[5] Luke Qi, Ali Khalatpour, Jason F Herrmann, **Taewon Park**, Devin Dean, Sam Robison, Alexander Hwang, Hubert Stokowski, Darwin Serkland, Martin M Fejer, Amir H Safavi-Naeini. "Low-loss, highly tunable Sagnac loop reflectors and Fabry–Pérot cavities on thin-film lithium niobate." *Optics Letters* 50(16), 5173–5176 (2025).

[6] Hubert S Stokowski, Devin J Dean, Alexander Y Hwang, **Taewon Park**, Oguz Tolga Celik, Timothy P McKenna, Marc Jankowski, Carsten Langrock, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Integrated frequency-modulated optical parametric oscillator." *Nature* 627(8002), 95–100 (2024).

[7] Oguz Tolga Celik, Nancy Yousry Ammar, **Taewon Park**, Hubert S Stokowski, Kevin KS Multani, Alexander Y Hwang, Samuel Gyger, Yudan Guo, Martin M Fejer, Amir H Safavi-Naeini. "Roles of temperature, materials, and domain inversion in high-performance, low-bias-drift thin film lithium niobate blue light modulators." *Optics Express* 32(21), 36160–36170 (2024).

[8] Sunil Pai, Zhanghao Sun, Tyler W Hughes, **Taewon Park**, Ben Bartlett, Ian A D Williamson, Momchil Minkov, Maziyar Milanizadeh, Nathnael Abebe, Francesco Morichetti, Andrea Melloni, Shanhui Fan, Olav Solgaard, David A B Miller. "Experimentally realized in situ backpropagation for deep learning in photonic neural networks." *Science* 380(6643), 398–404 (2023).

[9] Sunil Pai, **Taewon Park**, Marshall Ball, Bogdan Penkovsky, Michael Dubrovsky, Nathnael Abebe, Maziyar Milanizadeh, Francesco Morichetti, Andrea Melloni, Shanhui Fan, Olav Solgaard, David A B Miller. "Experimental evaluation of digitally verifiable photonic computing for blockchain and cryptocurrency." *Optica* 10(5), 552–560 (2023).

[10] Hubert S Stokowski, Timothy P McKenna, **Taewon Park**, Alexander Y Hwang, Devin J Dean, Oguz Tolga Celik, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Integrated quantum optical phase sensor in thin film lithium niobate." *Nature Communications* 14(1), 3355 (2023).

[11] Alexander Y Hwang, Hubert S Stokowski, **Taewon Park**, Marc Jankowski, Timothy P McKenna, Carsten Langrock, Jatadhari Mishra, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Mid-infrared spectroscopy with a broadly tunable thin-film lithium niobate optical parametric oscillator." *Optica* 10(11), 1535–1542 (2023).

[12] Sunil Pai, Carson Valdez, **Taewon Park**, Maziyar Milanizadeh, Francesco Morichetti, Andrea Melloni, Shanhui Fan, Olav Solgaard, David A B Miller. "Power monitoring in a feedforward photonic network using two output detectors." *Nanophotonics* 12(5), 985–991 (2023).

Conference Publications

[13] **Taewon Park**, Sam Robison, Devin J Dean, Luke Qi, Oguz T Celik, Hubert S Stokowski, Martin M Fejer, Amir H Safavi-Naeini. "Integrated Optical Parametric Oscillator Squeezer on Thin-Film Lithium Niobate." *Nonlinear Optics*, M2A-5 (2025).

[14] Devin J Dean, **Taewon Park**, Hubert S Stokowski, Luke Qi, Sam Robison, Alexander Y Hwang, Jason Herrmann, Martin M Fejer, Amir H Safavi-Naeini. "Towards Efficient Optical Parametric Amplification on Chip by Second Harmonic Resonance." *Nonlinear Optics*, M2A-4 (2025).

[15] Hubert S Stokowski, Devin J Dean, Alexander Y Hwang, **Taewon Park**, Oguz Tolga Celik, Timothy P McKenna, Marc Jankowski, Carsten Langrock, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Frequency-Modulated Optical Parametric Oscillator in Thin Film Lithium Niobate." *CLEO: Science and Innovations*, STh3I-2 (2024).

[16] Alexander Y Hwang, Hubert S Stokowski, **Taewon Park**, Marc Jankowski, Timothy P McKenna, Carsten Langrock, Jatadhari Mishra, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Integrated Optical Parametric Oscillators for Mid-Infrared Spectroscopy." *Optical Sensors*, STh1C-2 (2024).

[17] **Taewon Park**, Hubert S Stokowski, Timothy P McKenna, Alexander Y Hwang, Devin J Dean, Luke Qi, Oguz Tolga Celik, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Squeezed light generation and analysis using a sub-threshold optical parametric oscillator on integrated thin-film lithium niobate photonic circuit." *CLEO: Science and Innovations*, SF1K-7 (2023).

[18] Hubert S Stokowski, Timothy P McKenna, **Taewon Park**, Alexander Y Hwang, Devin Dean, Oguz Tolga Celik, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Quantum-Enhanced Phase Detection in a Lithium Niobate Photonic Integrated Circuit." *CLEO: Fundamental Science*, FF2L-3 (2023).

[19] Hope Lee, Daniel Riedel, Jakob Grzesik, Jason F Herrmann, Shahriar Aghaeimeibodi, Jean-Michel Borit, Vahid Ansari, Hubert S Stokowski, Luke Qi, **Taewon Park**, et al. "Heterogeneous Integration of SiV-Centers in Diamond with Lithium Niobate Photonics for Quantum Networks." *2023 Conference on Lasers and Electro-Optics (CLEO)*, 1–2 (2023).

[20] Alexander Y Hwang, Hubert S Stokowski, **Taewon Park**, Marc Jankowski, Timothy P McKenna, Jatadhari Mishra, Martin M Fejer, Amir H Safavi-Naeini. "Singly-Resonant Mid-IR Optical Parametric Oscillator in Lithium Niobate Nanophotonics." *2023 Conference on Lasers and Electro-Optics (CLEO)*, 1–2 (2023).

[21] Hope Lee, Daniel Riedel, Jason Herrmann, Yakub Grzesik, Jean-Michel Borit, Vahid Ansari, Hubert S Stokowski, Luke Qi, **Taewon Park**, Alexander Y Hwang, et al. "Heterogeneous Integration of Diamond and Lithium Niobate Nanophotonics for Quantum Networks." *APS March Meeting Abstracts* (2023).

[22] Luke Qi, Hubert S Stokowski, Timothy P McKenna, Jason Herrmann, **Taewon Park**, Alexander Y Hwang, Vahid Ansari, Martin M Fejer, Jelena Vučković, Amir H Safavi-Naeini. "Cryogenic Visible-to-Infrared Quantum Interface in a Lithium Niobate Photonic Circuit." *CLEO: Applications and Technology*, JTu2A-94 (2023).

[23] Hubert S Stokowski, Devin J Dean, Alexander Y Hwang, **Taewon Park**, Oguz Tolga Celik, Marc Jankowski, Vahid Ansari, Martin M Fejer, Amir H Safavi-Naeini. "Optical Frequency Comb via Electro-Optically Locked Parametric Oscillator in Thin Film Lithium Niobate." *Nonlinear Optics*, Tu1A-5 (2023).

[24] Devin J Dean, **Taewon Park**, Hubert S Stokowski, Alexander Y Hwang, Luke Qi, Martin M Fejer, Amir H Safavi-Naeini. "Efficient Second Harmonic Resonant Nonlinear Device in Thin-Film Lithium Niobate." *Nonlinear Optics*, W2B-6 (2023).

[25] Alexander Y Hwang, Hubert S Stokowski, **Taewon Park**, Marc Jankowski, Timothy P McKenna, Jatadhari Mishra, Martin M Fejer, Amir H Safavi-Naeini. "Integrated Lithium Niobate OPO for Tunable Mid-IR Spectroscopy." *Nonlinear Optics*, W2B-7 (2023).

[26] Oguz Tolga Celik, Nancy Yousry Ammar, Hubert S Stokowski, **Taewon Park**, Amir Safavi-Naeini. "Visible electro-optic modulator array in thin-film lithium niobate for atomic qubit control." *Frontiers in Optics*, FM5B-2 (2023).

[27] Isaac Luntadila Lufungula, Felix M Mayor, Jason F Herrmann, **Taewon Park**, Hubert S Stokowski, Alexander Y Hwang, Camiel Op De Beeck, Okan Atalar, Wentao Jiang, Bart Kuyken, et al. "Tunable dual wavelength laser on thin film lithium niobate." *2023 IEEE Photonics Conference (IPC)*, 1–2 (2023).

[28] Oguz Tolga Celik, Nancy Yousry Ammar, Hubert S Stokowski, **Taewon Park**, Amir H Safavi-Naeini. "Bias-stable Sub-Volt Visible Electro-optic Modulator in Thin-Film Lithium Niobate." *2023 IEEE Photonics Conference (IPC)*, 1–2 (2023).

[29] **Taewon Park**, Hubert S Stokowski, Timothy P McKenna, Alexander Y Hwang, Vahid Ansari, M M Fejer, Amir H Safavi-Naeini. "Second harmonic generation of blue light on integrated thin-film lithium niobate waveguides." *2022 Conference on Lasers and Electro-Optics (CLEO)*, 1–2 (2022).

[30] Sunil Pai, **Taewon Park**, Bogdan Penkovsky, Maziyar Milanizadeh, Marshall Ball, Michael Dubrovsky, Nathnael Abebe, Francesco Morichetti, Andrea Melloni, Olav Solgaard, et al. "Lighthash: Experimental evaluation of a photonic cryptocurrency." *2022 Conference on Lasers and Electro-Optics (CLEO)*, 1–2 (2022).

[31] Sunil Pai, Tyler W Hughes, **Taewon Park**, Ben Bartlett, Ian Williamson, Momchil Minkov, Maziyar Milanizadeh, Nathnael Abebe, Francesco Morichetti, Andrea Melloni, et al. "Inference and Gradient Measurement for Backpropagation in Photonic Neural Networks." *2022 Conference on Lasers and Electro-Optics (CLEO)*, 1–2 (2022).

[32] **Taewon Park**, Youngjae Jeong, Kyoungsik Yu. "Construction of a Multi-Wavelength Unitary Operator via Cascaded Optical Resonators." *CLEO: QELS_Fundamental Science*, JTh2B-5 (2020).

Presentations

- **August 2025**: Nonlinear Optics Conference. *"Integrated Optical Parametric Oscillator Squeezer on Thin-Film Lithium Niobate"*, Hawaii.
- **July 2024 (Invited talk)**: Workshop on Quantum Light Generation, Application, and Detection. *"Single-mode squeezed-light generation and tomography with an integrated optical parametric oscillator"*, JILA, University of Colorado, Boulder, CO, USA.
- **August 2024**: NTT Coherent Ising Machine (CIM) Consortium. *"Towards Integrated Quantum Squeezed Light Sensor"*, Virtual.
- **July 2023**: Gordon Research Conference – Quantum Sensing (GRC). *"Squeezed light from a photonic integrated circuit"*, Les Diablerets, VD, Switzerland.
- **May 2023**: Conference on Lasers and Electro-Optics (CLEO). *"Squeezed light generation and analysis using a sub-threshold OPO on integrated thin-film lithium niobate photonic circuit"*, San Jose, CA, USA.
- **May 2022**: Conference on Lasers and Electro-Optics (CLEO). *"Second harmonic generation of blue light on thin-film lithium niobate waveguides"*, San Jose, CA, USA.
- **May 2020**: Conference on Lasers and Electro-Optics (CLEO). *"Construction of a multi-wavelength unitary operator via cascaded optical resonators"*, San Jose, CA (virtual), USA.
- **August 2019**: SPIE Optics + Photonics. *"Compact multi-wavelength universal linear optical component based on cascaded resonators"*, San Diego, CA, USA.

Professional Experience and Awards

- **2025** Best Poster Award (runner-up), DARPA INSPIRED Program PI Meeting
- **2020–2021** Stanford Department Fellowship, Stanford University
- **2020** Research Assistant, Department of Physics, Seoul National University (Advisor: Dohun Kim)
- **2020** Incubic/Milton Chang Travel Grant for CLEO 2020, CLEO

- **2020** Fulbright Graduate Study Award, Declined
- **2020** Graduation with Honors, KAIST
- **2019** Research Excellence Award for SPIE Optics+Photonics Conference, Sponsor: MKS Instruments
- **2019** Excellent Award for Undergraduate Research Program, KAIST
- **2019** Dean's List for Outstanding GPA, KAIST
- **2018–2019** Undergraduate Research Assistant, Electrical Engineering, KAIST (Advisor: Kyoungsik Yu)
- **2018, 2019** Undergraduate Research Program Scholarship, KAIST
- **2018** Undergraduate Research Student, Electrical Engineering, KAIST (Advisor: Hyunjoo Jenny Lee)
- **2013–2018** KAIST Scholarship (Merit-based scholarship for 7 semesters), Korea Student Aid Foundation

In the Media

- **Stanford News**: "Energy-efficient optical amplifier for biosensing and data communication" (2026)
- **Stanford News**: "Low-power, high-precision measurement tool could boost tech" (2024)
- **Optics & Photonics News, OPTICA**: "Energy-Efficient Bitcoin Mining Through Light" (2023)